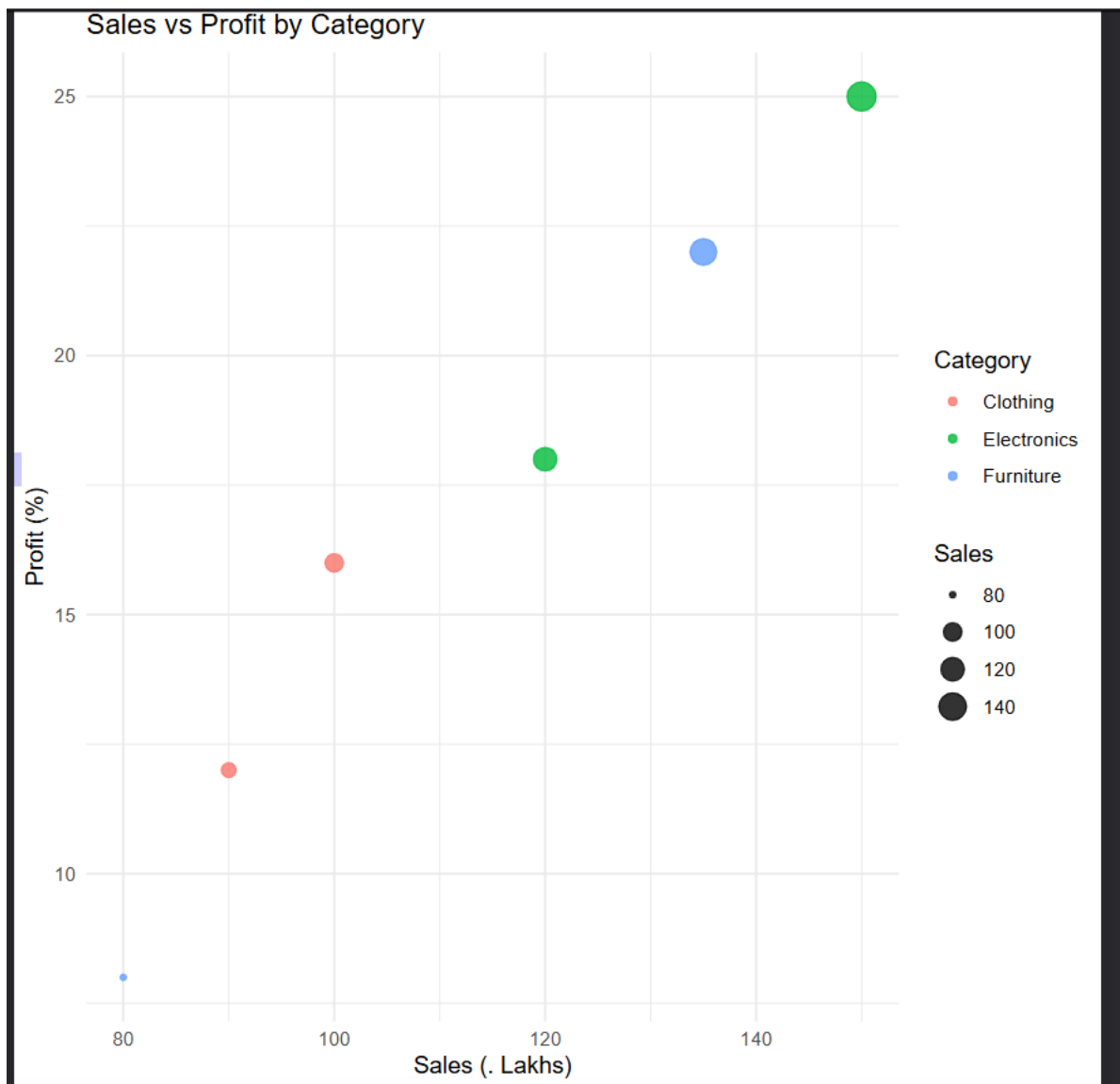


Question -1

(i)



(ii)

Mapping of Aesthetics

- **X-axis:** Sales (₹ Lakhs)
- **Y-axis:** Profit (%)
- **Color:** Category (Electronics, Clothing, Furniture)
- **Size:** Sales (Higher sales appear as larger points)

(iii) Interpretation of the Relationship

- There is a **positive relationship** between **Sales** and **Profit**.
- Products with higher sales generally have higher profit percentages.
- **P3** has the highest sales (₹150 Lakhs) and the highest profit (25%).
- **P4** has the lowest sales (₹80 Lakhs) and the lowest profit (8%).
- Electronics products tend to perform better than Clothing and Furniture in this dataset.

(iv)

Variable	Data Type	Scale Used
Sales	Numerical (Continuous)	Continuous Scale
Profit	Numerical (Continuous)	Continuous Scale
Category	Categorical	Discrete Color Scale

v. Alternative Visualization

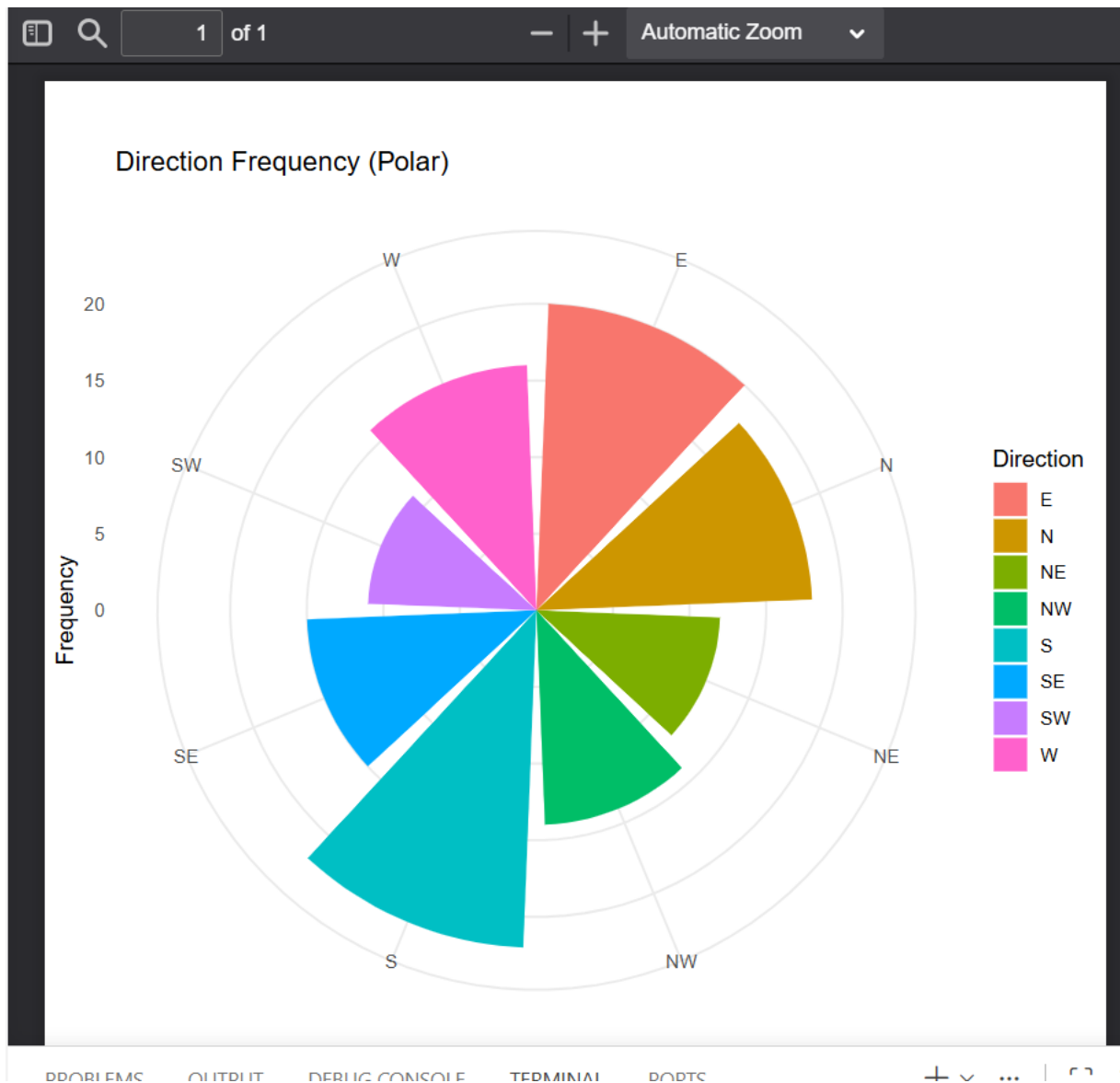
A **Bubble Chart** (larger bubbles for higher sales and different colors for categories) is a good alternative because it clearly shows:

- Sales
- Profit
- Category

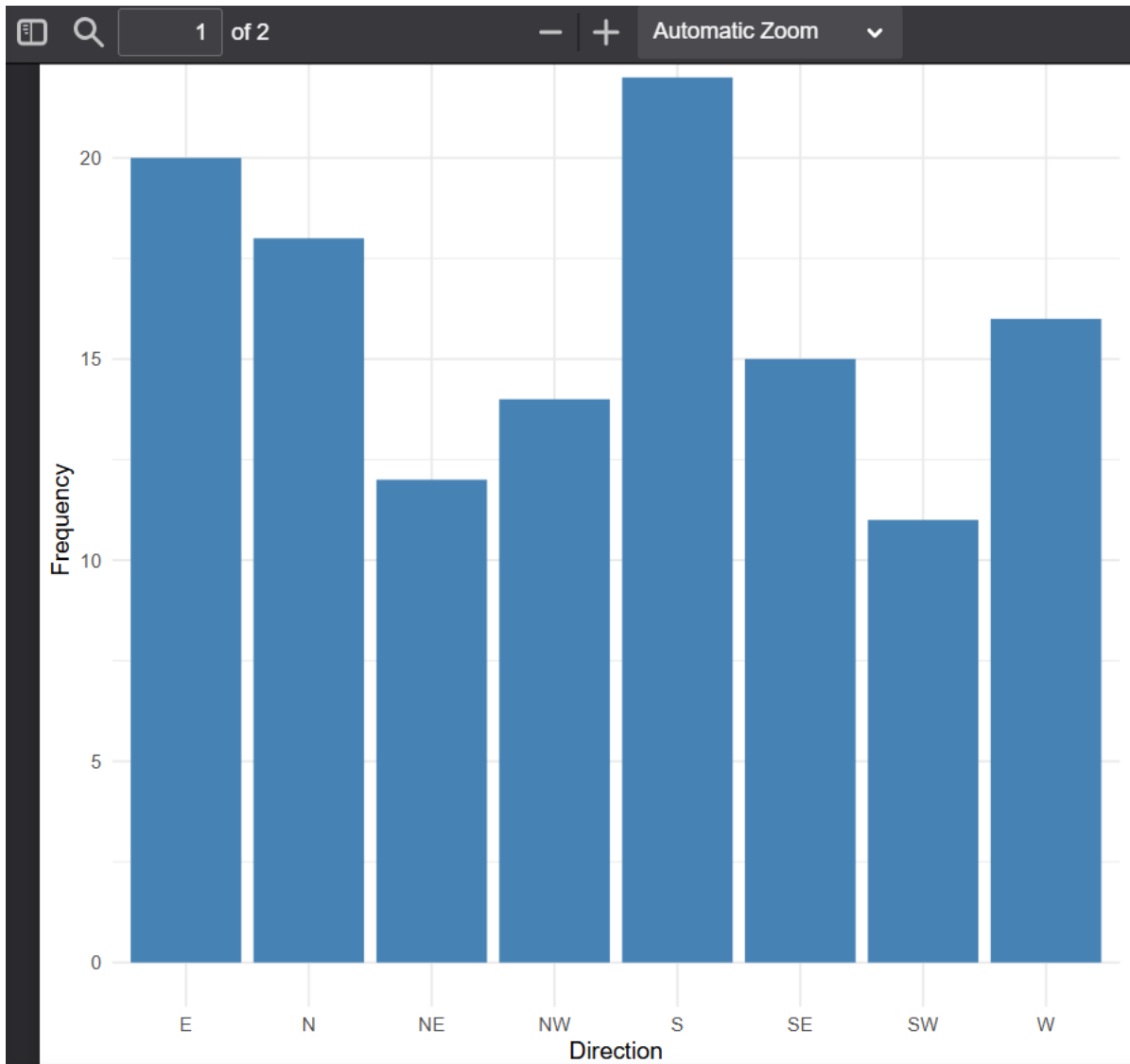
Another suitable option is a **Grouped Bar Chart** comparing Sales and Profit across products.

Question-2

(i)



(ii)



(iii)

Compare the effectiveness of both visualizations.

- **Cartesian chart** is better for comparing exact frequencies.
- **Polar chart** is better for showing directional or circular patterns.

(iv)

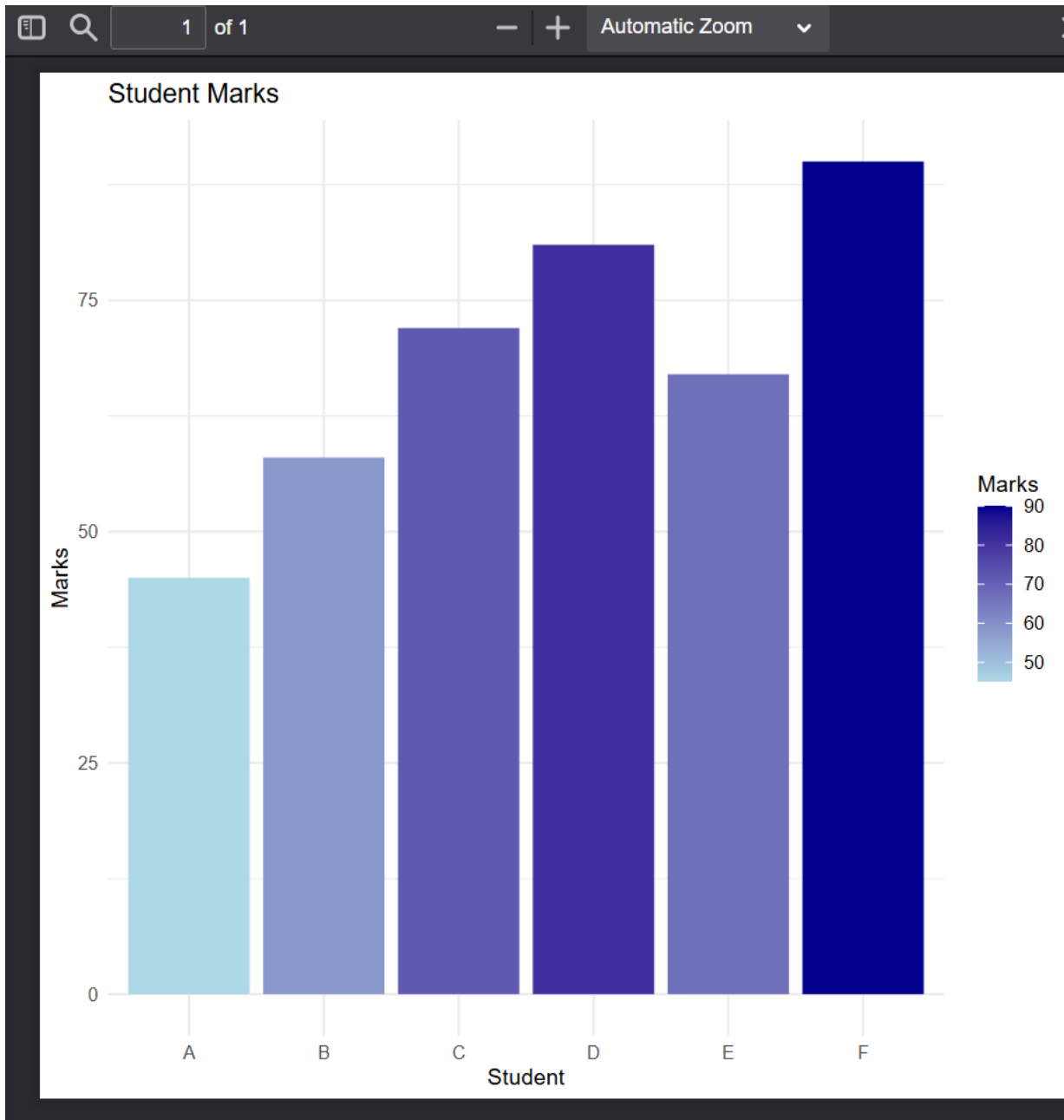
Which coordinate system is more appropriate for directional data?

Polar coordinate system is more appropriate because directional data is naturally represented in a circular format.

(v)

The Polar coordinate system arranges directions (N, NE, E, SE, S, SW, W, NW) around a circle, making it easier to understand directional relationships. The Cartesian system is better for comparing values but does not effectively represent direction.

Question-3



(ii)

Identify the students with the highest and lowest marks.

- **Highest Marks:** Student **F (90)**
- **Lowest Marks:** Student **A (45)**

(iii)

Why is a sequential color scale appropriate?

A **sequential color scale** is appropriate because **Marks are continuous numerical values**. It uses lighter colors for lower marks and darker colors for higher marks, making differences easy to understand.

(iv)

Suggest a color palette suitable for color-blind users.

A good color-blind-friendly palette is **Viridis**.

Example in R:

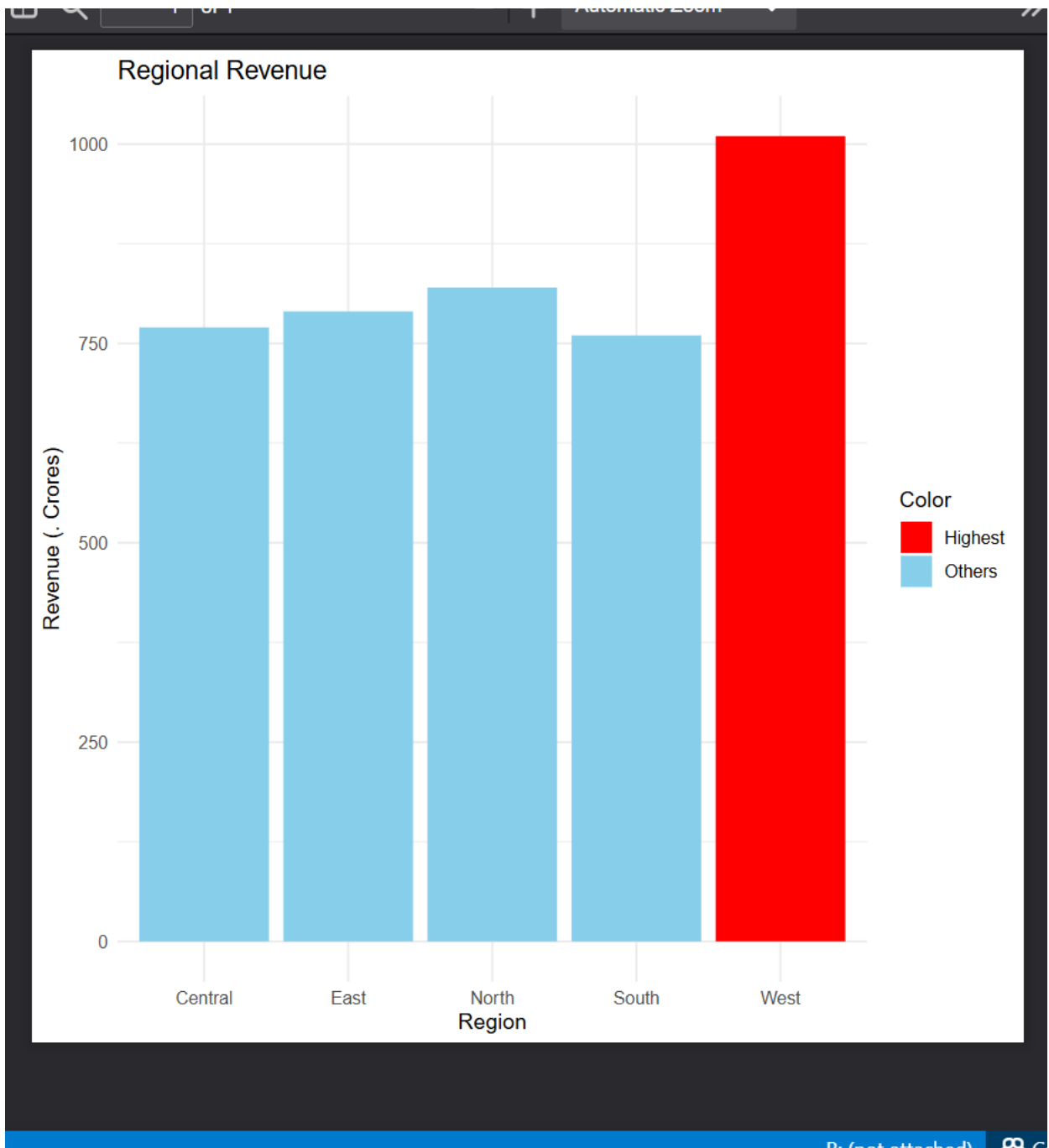
```
scale_fill_viridis_c()
```

(v)

Difference between sequential and categorical color scales.

- **Sequential color scale:** Used for **continuous numerical data** (e.g., Marks, Temperature, Sales).
- **Categorical color scale:** Used for **categorical data** (e.g., Gender, Department, Category), where each category gets a different color.

Question-4



(ii)

Explain how highlighting improves data interpretation.

Highlighting draws attention to the most important data, making it easier to identify key values and compare them quickly.

(iii)

Identify the highest-performing region.

West is the highest-performing region with **₹1010 Crores** revenue.

(iv)

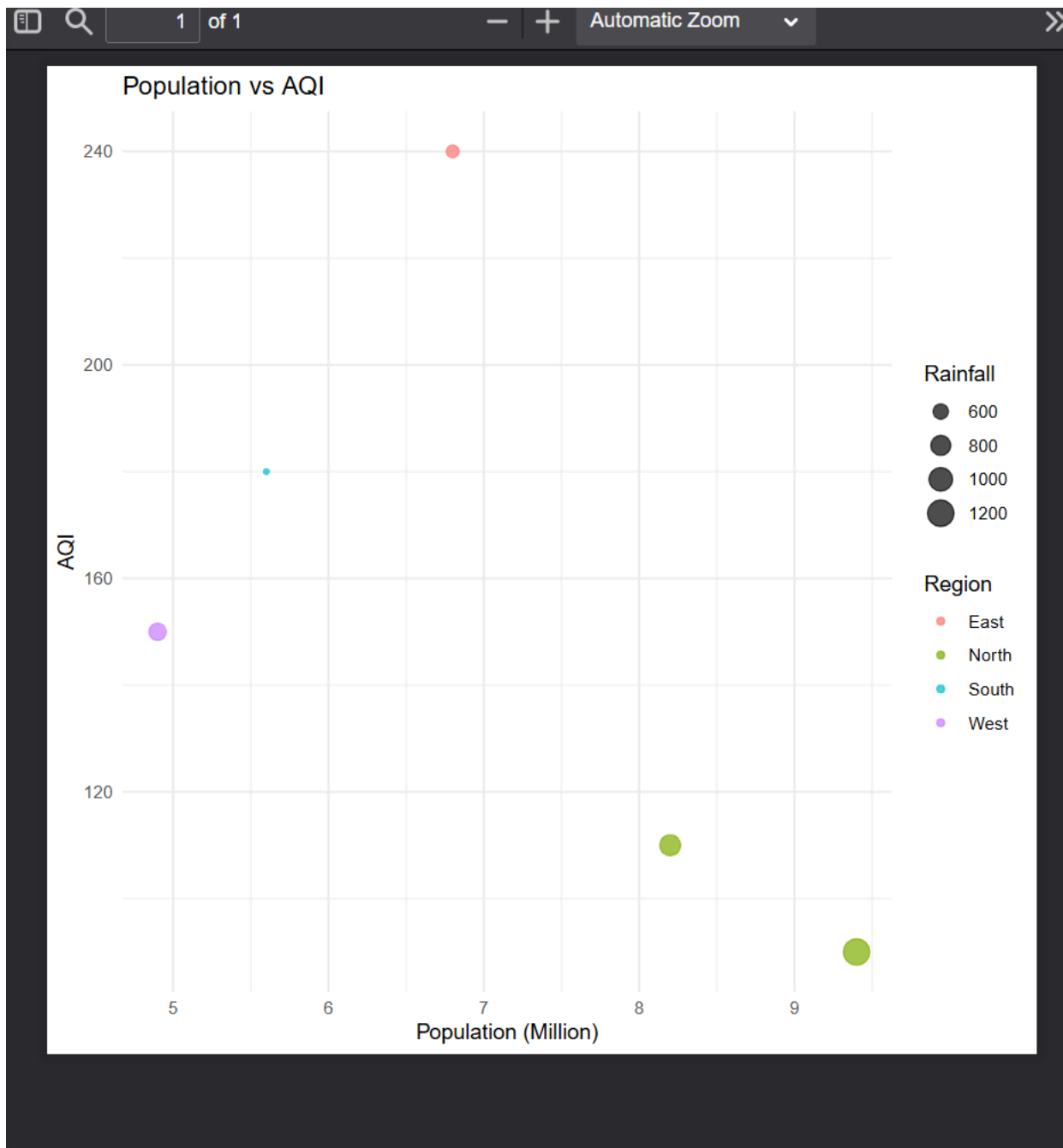
Discuss how excessive use of colors can affect interpretation.

Using too many colors can confuse viewers, reduce readability, and make it difficult to identify the most important information.

(v)

uggest one alternative highlighting technique other than color.

Use **data labels (annotations)** to highlight important values. Other acceptable techniques include **larger bar size**, **bold outlines**, or **arrows**.



(ii)

Suitable aesthetics

- **X-axis:** Population
- **Y-axis:** AQI
- **Size:** Rainfall
- **Color:** Region

(iii) Interpret the relationship between Population and AQI.

There is **no clear relationship** between Population and AQI. Some cities with higher population have lower AQI, while others with lower population have higher AQI.

(iv) Identify the continuous and categorical variables.

Continuous variables:

- Population
- AQI
- Rainfall

Categorical variables:

- City
 - Region
-

(v) Recommend one improvement to enhance readability.

Add **data labels** (city names) to each bubble so that each city can be identified easily.